稻济天下

Rice for the World



杂交稻的丰收，怎样由许多人共同促成？

READ 纯净阅读

CHECK 中文核对

LEARN 词汇记忆

RECALL 回想巩固

英文原文

In July 1964, inspecting rice ear by ear in a paddy at Anjiang Agricultural School in Hunan province, the instructor Yuan Longping spotted a peculiar plant. It was male-sterile, shedding no viable pollen. Rice pollinates itself, so breeding hybrid seed in bulk depends on such plants. In February 1966 he detailed the anomaly in the journal Chinese Science Bulletin.

The specimen launched a six-year hunt: Yuan and his colleagues could not breed a sterile line whose offspring reliably inherited the trait. The breakthrough came on 23 November 1970, in a marsh near Nanhong Farm in Sanya, Hainan. Yuan's assistant, Li Bihu, and Feng Keshan, a farm technician, found wild rice whose pollen was abortive.

Then came Yuan's decisive move: rather than withhold the strain, he distributed it to collaborators nationwide. Hybrids need three distinct lines: a sterile line, one to maintain its sterility, and one to restore fertility in the hybrids. Yan Longan's team in Jiangxi bred sterile lines and Zhang Xiancheng in Guangxi first identified a restorer line. In October 1973, at a national conference in Suzhou, Yuan declared the three-line system complete.

By 1976 hybrids were under cultivation on about 140,000 hectares. The World Food Prize Foundation estimates that hybrid rice can yield up to a fifth more grain than conventional varieties. By the early 2000s it covered almost half of China's rice fields.

Yuan, who shared the 2004 World Food Prize with Sierra Leone's Monty Jones, died in Changsha on 22 May 2021, aged 91. Yet popular myth portrays him as a solitary genius. His eye began the search; his generosity let many hands bring in the harvest.

中文译文

1964年7月，湖南安江农校教员袁隆平在稻田里逐穗检查，发现一株不同寻常的水稻：它雄性不育，无法散出有活性的花粉。水稻通常自花授粉，因此大批量培育杂交种子，需要这样的植株。1966年2月，他在《科学通报》详细报告了这一反常现象。

这株标本开启了长达六年的搜寻。袁隆平与同事一直无法育成一个不育系，使其后代稳定继承雄性不育的性状。突破出现在1970年11月23日：在海南三亚南红农场附近的沼泽中，袁隆平的助手李必湖和农场技术员冯克珊找到了花粉败育的野生稻。

袁隆平随后作出了关键决定：他没有将这个品系留在自己手中，而是把材料分给全国各地的协作者。杂交稻需要三个不同的系：不育系、保持其不育性的保持系，以及让杂交后代恢复育性的恢复系。江西颜龙安团队育成了不育系，广西张先程率先找到了恢复系。1973年10月，袁隆平在苏州的一次全国会议上宣布，三系已经配套。

到1976年，杂交稻种植面积约为14万公顷。世界粮食奖基金会估计，它比常规品种最多可多产五分之一的稻谷。到21世纪初，杂交稻已覆盖中国近一半稻田。

袁隆平与塞拉利昂的蒙蒂·琼斯共同获得2004年世界粮食奖。他于2021年5月22日在长沙去世，享年91岁。然而，流行的传奇常把他描绘成单枪匹马的天才。他的敏锐观察开启了搜寻，他愿意分享材料，才让许多人的共同努力促成丰收。

雄性不育指缺少有效花粉；植株仍能作母本接受其他植株的花粉。文中的增产两成是基金会估计的最高幅度。

核心词汇

先记这22项。连着下方的原文短语一起回想。

inspect v.
仔细检查
inspecting rice ear by ear

peculiar adj.
不同寻常的
a peculiar plant

sterile adj.
不育的；本文指雄性不育
It was male-sterile

viable adj.
有活性的；能正常发育的
no viable pollen

pollen n.
花粉
no viable pollen

hybrid adj.
杂交的
breeding hybrid seed in bulk

in bulk 介词短语
大批量地
breeding hybrid seed in bulk

anomaly n.
反常现象
detailed the anomaly

specimen n.
标本
The specimen launched a six-year hunt

breed v.
培育；繁育
breed a sterile line

inherit v.
遗传获得
reliably inherited the trait

trait n.
性状；特征
reliably inherited the trait

breakthrough n.
突破
The breakthrough came

abortive adj.
（花粉）败育的
whose pollen was abortive

decisive adj.
关键的；决定性的
Yuan's decisive move

withhold v.
不予提供；保留
rather than withhold the strain

collaborator n.
协作者
collaborators nationwide

distinct adj.
不同的
three distinct lines

restore v.
恢复
restore fertility in the hybrids

fertility n.
育性；繁殖能力
restore fertility in the hybrids

yield v.
产出
yield up to a fifth more grain

generosity n.
慷慨；乐于分享
his generosity

文中词汇与搭配

补充词汇共37项。保留每个表达在本文中的意思。

paddy n.
稻田

instructor n.
教员

shed v.
散出；释放

pollinate itself 动词短语
自花授粉

detail v.
详细描述

journal n.
学术期刊

launch v.
开启；发起

hunt n.
搜寻

colleague n.
同事

offspring n.
后代

reliably adv.
稳定可靠地

marsh n.
沼泽

technician n.
技术员

strain n.
品系

distribute v.
分发

to prep.
给；向

nationwide adv.
在全国范围

maintain v.
保持

文中词汇与搭配 续

补充词汇共37项。保留每个表达在本文中的意思。

sterility n.
不育性

identify v.
找出；确认

national conference n.
全国性会议

declare v.
宣布

under cultivation 介词短语
处于种植中

hectare n.
公顷

estimate v.
估计

grain n.
谷物；此处指稻谷

conventional adj.
常规的

variety n.
品种

cover v.
覆盖；占据

rice field n.
稻田

World Food Prize n.
世界粮食奖

myth n.
流传的错误观念

portray v.
描绘

as prep.
作为；描绘成

solitary adj.
独自一人的

genius n.
天才

bring in the harvest 动词短语
收获庄稼；促成丰收

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 In July 1964, inspecting rice ear by ear in a paddy at Anjiang Agricultural School in Hunan province, the instructor Yuan Longping spotted a _____ plant.

02 It was male-sterile, shedding no _____ pollen.

03 Then came Yuan's decisive move: rather than _____ the strain, he distributed it to collaborators nationwide.

04 The World Food Prize Foundation estimates that hybrid rice can _____ up to a fifth more grain than conventional varieties.

文章主线

- 难题 水稻通常自花授粉，大批杂交制种要用不产有效花粉的母株；袁隆平1964年发现这样的稻株，却难让后代稳定保持不育性。
- 协作 1970年李必湖、冯克珊找到花粉败育野生稻；袁隆平向全国团队共享材料，颜龙安团队育成不育系，张先程率先找到恢复系。
- 机制 1973年三系配套：不育系作母本；保持系为母本授粉，繁殖仍不育的母株；恢复系为母本授粉，产生可结实的杂交后代。
- 推广 1976年已种约14万公顷，21世纪初覆盖中国近半稻田；世界粮食奖基金会估计，最多比常规稻增产两成。
- 贡献 袁隆平的发现、材料共享与全国协作共同促成丰收，并非一人独创；他2004年共获世界粮食奖，2021年去世。

参考答案　1 peculiar 2 viable 3 withhold 4 yield